

3.7 Nuclear Pollution

An atom is characterised by its atomic number and atomic mass. Carbon is one of the most frequently occurring chemical in living materials which is not radioactive. It has exactly 12 atomic mass units. Other atoms are measured in relation to Carbon-12. Carbon-14 has two extra neutrons in its nucleus has the ability of radioactive.

Hydrogen, has an atomic number of 1 and an atomic mass of 1. Isotopes of hydrogen have the same atomic number but a higher atomic mass. Hydrogen-2 or deuterium is an isotope of hydrogen. It has an atomic number of 1 and an atomic mass of 2. It is not radioactive. The increased atomic mass is due to an added neutron in the nucleus. Deuterium is in the 'heavy water' used in the nuclear reactors. Hydrogen-3, called tritium has the ability of radioactivity. It consists of two neutrons and a proton in the nucleus. It is produced in a nuclear reaction.

When radium 226 decays, it loses a positively charged alpha particle from its nucleus. An alpha particle has two protons and a mass of 4 atomic units. Loss of the alpha particle changes radium 226 into another element, called radon 222. Radium 226 is a radioactive solid under normal conditions, whereas radon 222 is a radioactive gas. Loss of one or more protons changes the chemical element into a different chemical. Absorption or loss of a neutron gives an isotope of the same chemical since chemical properties are determined by the number of protons and electrons in an atom.

Nuclear reactors are used to generate power by using nuclear fuel. Uranium and its isotopes are mostly used in all the nuclear reactors. Generally, Uranium-235 and Uranium-238 are used in the nuclear reactors for power generation. Ur-235 alone and along with Ur-238 are used in light-water reactors and heavy-water reactors respectively.

Some Important Terms

1. **Alpha particle** is an electrically charged (+) particle emitted from the nucleus of some radioactive chemicals, eg. plutonium. It contains 2 protons and 2 neutrons, and is the largest of the atomic particles emitted by radioactive chemicals. It can cause ionisation.
2. **Beta particle** is an electrically charged (-) particle emitted from some radioactive chemicals. It has the mass of an electron. Krypton 85, emitted from nuclear power plants, is a strong beta emitter. Beta particles can cause ionisation.
3. **Curie** is a measure of radioactivity. One curie equals 3.7×10^{10} nuclear transformations per second. Ci is the symbol used.
 - (a) **Microcurie**: one-millionth of a curie. (3.7×10^4 disintegrations per second) mCi is the symbol used.
 - (b) **Picocurie**: one-trillionth (10^{-12}) of a curie. pCi is the symbol used. One pCi is equal to the decay of about two radioactive atoms per minute.
4. **Dose** is the energy imparted to matter by nuclear transformations (radioactivity).
 - (a) **Rad** = 100 ergs per gram.
1 GRAY = 100 rad = 10,000 ergs per gram.
 - (b) **Rem** = rads $\times$ Q where Q is a quality factor which attempts to convert rads from different types of radioactivity into a common scale of biological damage.
1 SIEVERT = 100 rad.
5. **Gamma rays** are short wave-length electromagnetic radiation released by some nuclear transformations. It is similar to X-ray and will penetrate through the human body. Iodine 131 emits gamma rays. Both gamma and X-rays cause ionisation.
6. **Ionisation** Sufficient energy is deposited in a neutral molecule to displace an electron, thus replacing the neutral molecule with positive and negative ions.

Sources of Nuclear Pollution

Initially the Uranium is extracted from ores. It is refined in chemical process and it is fabricated into fuel rods. These fuel rods are inserted into the nuclear reactor. Generally, the rods are kept in the reactor for a period of about two years. During that period, they become intensely radioactive due to the production of fission products and formation of elements like plutonium, americium and curium.

The vast majority of highly radioactive wastes are generated from commercial nuclear power reactors. Highly radioactive wastes contain two types of wastes.

1. Solid irradiated nuclear fuel assemblies (generally called “spent” or “used” fuel rods) and
2. Liquid high-level radioactive wastes resulting from the “reprocessing” (extraction of fissile plutonium and uranium) of solid irradiated fuel rods.

Irradiated nuclear fuel rods discharged from commercial nuclear power plants are highly radioactive, a million times more so than when they were first loaded into a reactor core as “fresh” fuel. If unshielded, irradiated nuclear fuel just removed from a reactor core could deliver a lethal dose of radiation to a person standing three feet away in just seconds. Even after decades of radioactive decay, a few minutes unshielded exposure could deliver a lethal dose. Certain radioactive elements (such as plutonium-239) in “spent” fuel will remain hazardous to humans and other living beings for hundreds of thousands of years. Other radioisotopes will remain hazardous for millions of years.

Highly radioactive wastes are dangerous and deadly wherever they are, whether stored at reactor sites (indoors in pools or outdoors in dry casks); transported on the roads, rails, or waterways; or dumped.

Low-level waste includes items that have become contaminated with radioactive material or have become radioactive through exposure to neutron radiation. Contaminated cloths, protective shoe covers, wiping rags, mops, filters, reactor water treatment residues, equipments and tools, luminous dials, medical tubes, swabs, injection needles, syringes, and laboratory animal carcasses and tissues are the typical examples of low level wastes.

Pollution from mining : Pollution created from mining of uranium is the important in considering the whole nuclear fuel-cycle. Near the uranium mining areas, the locally available rivers are seriously polluted due to poor tailings management. In due course, the situation has been largely overcome but there have still been occurrences of tailings-pond failures. In all mining operations this is a lifelong problem. This problem needs constant caution.

Mill tailings wastes are the residues remaining after the processing of natural ore to extract uranium and thorium. Disposal of dry uranium tailings causes release of radioactive radon gas but in amounts that would not add significantly to the exposures

of surrounding communities. Radon is constantly being released from all soil, especially when it is disturbed, e.g., by ploughing.

Nuclear Power Stations : Nuclear-electric power stations release radioactive materials to air and water in amounts small enough not to harm the environment. The most obvious environmental effects that nuclear reactors do *not* cause are those resulting from combustion.

Coal-fired plants : Coal-fired plants also emit more radiation. In certain cases, the radioactive emissions from a coal-fired plant can be as high as, or even higher than, those from a nuclear plant. The radioactive emission depends on source of the coal and its impurities. Very little amount of uranium and thorium are present in coal. At the time of decaying these elements produce radium and radon. These decaying elements are disposed into the environment either from the chimney of the plant or disposed through the ash. Also this radioactive element causes the ground water pollution by leaching.

Effects of Nuclear Pollution

The nuclear pollution affects both the biotic and abiotic components of the environment. Its effect depends on the nature of radiation, level of radioactivity and the extent of the radiation.

The low level radioactive materials pose no risk at all, but can still be detected by radiation instrumentation, thereby being more of an annoyance than a threat. In the case of low-level contamination by isotopes with a short half-life, the best course of action may be to simply allow the material to naturally decay. Longer-lived isotopes should be cleaned up and properly disposed of.

High level of radiation pose major risks to people and the environment. The high level radiation affects the living beings in both externally and internally. Radioactive contamination can enter the body through ingestion, inhalation, absorption, or injection. For this reason, it is important to use personal protective equipment when working with radioactive materials. Radioactive contamination may also be ingested as the result of eating contaminated plants and animals or drinking contaminated water or milk from exposed animals.

Detection and Analysis of Radioactive Wastes

Detection of radiation is generally carried out by using devices designed to measure the radiation dose. The absorbed dose is defined as the energy imparted to matter by the ionizing radiation per unit mass of irradiated material at a given location. It is measured in terms of *rad*. Rads of γ and β are normally equivalent to rems, and are used interchangeably. The *sievert* (Sv) is the SI unit and equals 100 rems.

Some instruments used in the detection of radiation include:

- (i) Geiger Mueller Counter
- (ii) Ionization Chamber
- (iii) Scintillation Counter
- (iv) Film Badges
- (v) Thermoluminescent Dosimeter (TLD).

The Geiger Mueller Counter, Ionization Chamber and Scintillation Counter are used for detection of surface contamination. Film badges and TLDs are used for long term monitoring of exposure to workers. Gamma Ray Spectroscopy is also used for analysis of gamma radiations.

Table 6.11: Maximum Permissible Dose Equivalent for Occupational Exposure

	<i>Dose</i>	<i>Unit</i>
● Combined whole body exposure		
Prospective annual limit	5	rems/year
Retrospective annual limit	10–15	rems/year
Long-term accumulation	$(N-18) \times 5$	rems
● Skin	15	rems/year
● Hands	75	rems/year (25/qtr)
● Forearms	30	rems/year (10/qtr)
● Other organs, tissues and organ system	15	rems/year (5/qtr)
● Fertile women (with respect to fetus)	0.5	rems in gestation period
● Dose limits for the public, or occasionally exposed individuals		
Individual or occasional	0.5	rems/year
Students	0.1	rems/year
● Population dose limits		
Genetic	0.17	rems/year (average)
Somatic	0.17	rems/year (average)
● Emergency dose limits - Life saving		
Individual (older than 45, if possible)	100	rems
Hands and forearms	200	rems, additional (300 rems total)
● Emergency dose limits - Less urgent:		
Individual	25	rems
Hands and forearms	100	rems (total)
● Family of radioactive patients		
Individual (under age 45)	0.5	rems/year
Individual (over age 45)	5.0	rems/year

N is age in years

Source: NCRP (1975).

Classification of Radioactive Wastes

Radioactive wastes or Low-level Radioactive Wastes (LLRW) is a general term used for a wide range of materials contaminated with radioisotopes. LLRW may be disposed on specially designed landfills. These include wastes which are potential hazards and will persist long after such precautions as institutional controls, improved waste forms and deeper disposal have ceased to be effective. Low-level radioactive wastes are classified as Class A, B, and C wastes. An overview of these classes and their types and sources of generation are presented in Table 6.12.

Radioactivity in the environment comes from both natural and man-made sources. Natural sources are natural deposits of radioactive materials such as uranium and thorium. Although natural radioactivity is most likely to be encountered in the environment due to its widespread dispersal, man-made radioactivity poses the greatest environmental risk. Man-made sources include mining activities, nuclear power plants, medical and laboratory facilities, nuclear weapon testing etc.

Nuclear power plants are the main source of radioactive wastes. The process of energy recovery from nuclear fuel include mining of uranium containing ores, refining of uranium,

fabrication and use of fuel rods etc. All these activities produce low-level radioactive wastes. Hospitals, clinics, research laboratories etc. also generate these types of wastes, as the use of radioisotopes is rapidly becoming common e.g. in cancer treatments.

Table 6.12: Overview of Classes A, B, and C of Radioactive Wastes

<i>Characteristics</i>	<i>Class A</i>	<i>Class B</i>	<i>Class C</i>
Concentration	Low concentrations of radionuclides	Higher concentrations of radionuclides	Highest concentrations of radionuclides
Waste form	Does not require stabilization but may be stabilized	Requires stabilization for 300 years	Requires stabilization for 300 years or more
Intruder Protection	Decays to an acceptable levels Requires no additional measures to protect	After 100 years decays to acceptable levels, provided waste is recognizable Requires stabilization	After 500 years, decays to acceptable levels Required stabilization and deeper disposal or barriers
Segregation	Class A must be segregated from B and C	No need to segregate from Class C	No need to segregate from Class B
Waste Types	Protective clothing, paper, laboratory trash	Resins and filters from nuclear power plants	Reactor components high activity industrial wastes

Source: adapted from Liu and Liptak (1996).